Code No: 118AA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, May - 2019
ADHOC AND SENSOR NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	What is Multipoint Relay?	[2]
b)	What are the features of Cluster Head Gateway Switch routing protocol?	[3]
c)	Explain how to classify the Multicast routing protocol.	[2]
d)	What are the drawbacks of flooding?	[3]
e)	What is the motivation behind the design of protocols for WSN?	[2]
f)	What are the applications of WSNs?	[3]
g)	Explain about the integrity in adhoc networks?	[2]
h)	What are the assumptions of secure routing protocols in WSNs?	[3]
i)	Discuss about ns 2 simulator.	[2]
j)	What is TOSSIM?	[3]

PART - B

(50 Marks)

2.a) Explain how to create the mesh in ODMRP.
b) Distinguish between topology based and Position based approaches in adhoc networks. [5+5]

OR

3.a) What are the characteristics of MANETs?
b) List out the major forwarding strategies in Position Based routing. Explain the Greedy Packet Forwarding strategy. [5+5]

4.a) Explain why traditional TCP cannot be used in wireless adhoc networks.
b) Describe the Voronoi Diagram based Geocasting (VDG) protocol. [5+5]

OR

5.a) Explain the header format of TCP.
b) Explain the classification of broadcasting protocols in MANETs. Explain the Area based method. [5+5]

6.a) Explain the functional block diagram of sensor node.
b) Describe about the hierarchical Power Aware routing in WSNs. [5+5]

OR

7.a) What are the advantages of WSN over wired ones?
b) Explain how to find the Multiple Paths in WSNs. [5+5]

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8.a) Explain about the architecture of intrusion detection system for adhoc networks.
b) Explain the problems that affect secure adhoc routing. [5+5]

OR

9.a) Explain about the Octopus protocol.
b) Explain about the key management in sensor networks. [5+5]

10.a) Explain about the characteristics of nesC.
b) What is importance of TinyGALS? [5+5]

OR

11. Explain the architecture of TinyOS. [10]

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